

PROTOSPHERE -

Embedded Systems & IoT Workshop

Participant Hardware & Software Setup Guide

Please complete the software setup before the workshop so we can begin the hands-on activities without spending session time on installations.

1. What You Need

Category	Requirement
Hardware	Laptop with a working USB port
Hardware	USB-C data cable
Software	Arduino IDE
Software	ESP32 board package in Arduino IDE
Software	LiquidCrystal_I2C library
Account	GitHub account — recommended
Tool	Git — recommended

2. Install Arduino IDE

Download the official Arduino IDE: <https://www.arduino.cc/en/software/>

Bring Your Projects to Life with Arduino Software



The screenshot shows the Arduino IDE 2.3.10 download page. On the left is a preview of the IDE's code editor. On the right, the title 'Arduino IDE 2.3.10' is followed by a 'Release notes' link. Below this is a paragraph describing the new features: 'The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger. For more details, check the [Arduino IDE 2.0 documentation](#).' Below the text is a 'DOWNLOAD' button. A dropdown menu is open, showing various operating system and architecture options: 'Linux ZIP file (64-bit X86-64)', 'Windows Win 10 or newer (64-bit)', 'Windows MSI Installer', 'Windows ZIP file', 'Linux AppImage (64-bit X86-64)', 'Linux ZIP file (64-bit X86-64)' (which is selected with a checkmark), 'macOS Intel 12 Monterey or newer (64-bit)', and 'macOS Apple Silicon 12 Monterey or newer (64-bit)'.

Arduino IDE download page — choose the installer for your operating system.

After installation, open Arduino IDE once to confirm it launches correctly.

3. Install the ESP32 Board Package

1. Open Boards Manager

In Arduino IDE, go to Tools → Board → Boards Manager.

2. Search for ESP32

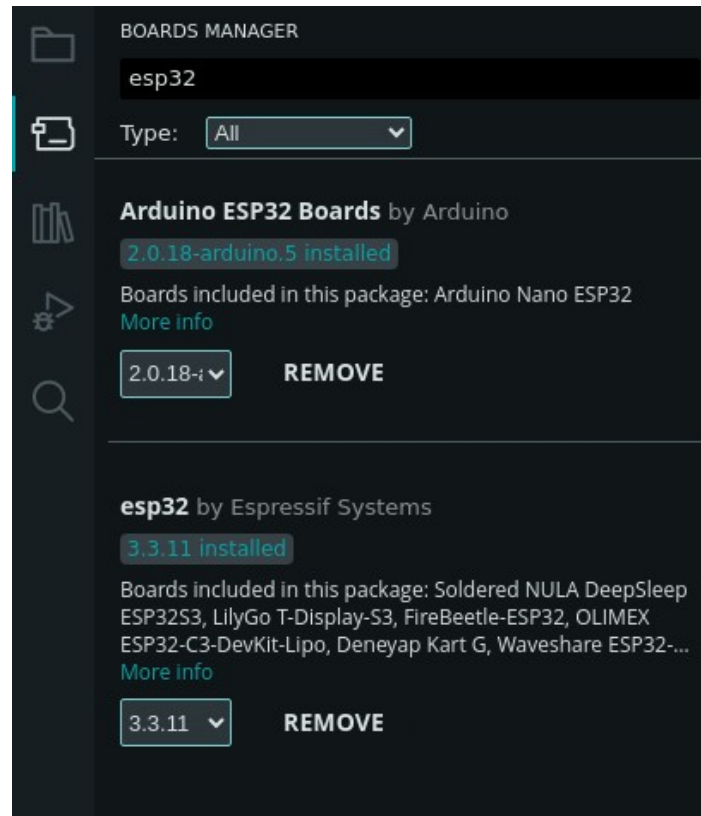
Search for “esp32”.

3. Install the Espressif package

Install “esp32 by Espressif Systems”.

4. Select your board

Use Tools → Board to select the ESP32 board provided for the workshop.



Boards Manager showing the ESP32 package by Espressif Systems.

Install the package from Espressif Systems.

4. Connect the ESP32

Connect the board using the USB-C data cable and confirm that its serial port appears under Tools → Port.

5. Install the I²C LCD Library

1. Open Library Manager

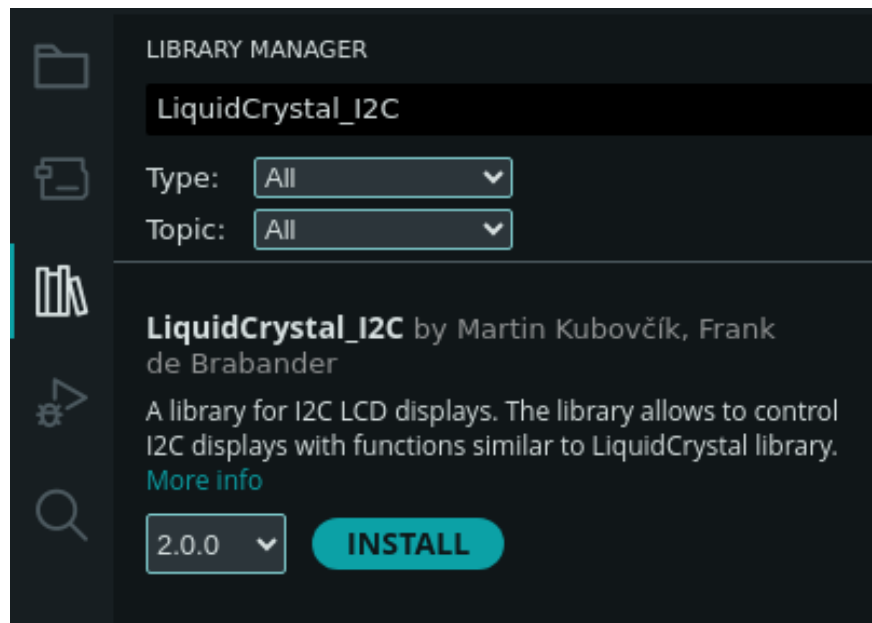
In Arduino IDE, go to Tools → Manage Libraries.

2. Search

Search for “LiquidCrystal_I2C”.

3. Install

Install the LiquidCrystal_I2C library shown in the Library Manager.



Library Manager search result for LiquidCrystal_I2C.

6. GitHub & Git (Recommended)

GitHub: Create a free account at <https://github.com/>

A GitHub account is recommended because workshop code and resources will be shared through a repository.

Git: Install Git from <https://git-scm.com/> if possible. It will be useful for downloading and managing workshop repositories.